

Lab 9 – Wireshark Analysis of 802.11 Wireless Network Protocol

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INFT1101: Wireless Network Security

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November 27, 2025

Task 1

Question 1: What are the SSIDs of the two access points that are issuing most of the beacon frames in this trace?

SSID of first access point: 30 Munroe St

SSID of second access point: linksys12

Question 2: What (in hexadecimal notation) is the source MAC address on the beacon frame from 30 Munroe St?

00:16:b6:f7:1d:51

```
▶ Frame 1: 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
▶ Radiotap Header v0, Length 24
▶ 802.11 radio information
▼ IEEE 802.11 Beacon frame, Flags: .....C
  Type/Subtype: Beacon frame (0x0008)
  ▶ Frame Control Field: 0x8000
    .000 0000 0000 0000 = Duration: 0 microseconds
  ▶ Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
  ▶ Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
  ▶ Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  ▶ Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  ▶ BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    .... .... 0000 = Fragment number: 0
    1011 0010 0110 .... = Sequence number: 2854
    Frame check sequence: 0x057e2608 [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: .....C]
▶ IEEE 802.11 Wireless Management
```

Question 3: What (in hexadecimal notation) is the destination MAC address on the beacon frame from 30 Munroe St?

ff:ff:ff:ff:ff:ff

```
▶ Frame 1: 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
▶ Radiotap Header v0, Length 24
▶ 802.11 radio information
▼ IEEE 802.11 Beacon frame, Flags: .....C
  Type/Subtype: Beacon frame (0x0008)
  ▶ Frame Control Field: 0x8000
    .000 0000 0000 0000 = Duration: 0 microseconds
  ▶ Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
  ▶ Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
  ▶ Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  ▶ Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  ▶ BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    .... .... 0000 = Fragment number: 0
    1011 0010 0110 .... = Sequence number: 2854
    Frame check sequence: 0x057e2608 [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: .....C]
▶ IEEE 802.11 Wireless Management
```

Question 4: What (in hexadecimal notation) is the MAC BSS id on the beacon frame from 30 Munroe St?

00:16:b6:f7:1d:51

```
‣ Frame 1: 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
‣ Radiotap Header v0, Length 24
‣ 802.11 radio information
‣ IEEE 802.11 Beacon frame, Flags: .....C
  Type/Subtype: Beacon frame (0x0008)
  ‣ Frame Control Field: 0x8000
    .000 0000 0000 0000 = Duration: 0 microseconds
  ‣ Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
  ‣ Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
  ‣ Transmitter address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  ‣ Source address: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
  ‣ BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)
    .... 0000 = Fragment number: 0
    1011 0010 0110 .... = Sequence number: 2854
    Frame check sequence: 0x057e2608 [unverified]
    [FCS Status: Unverified]
    [WLAN Flags: .....C]
‣ IEEE 802.11 Wireless Management
```

Question 5: The beacon frames from the 30 Munroe St access point advertise that the access point can support four data rates and eight additional “supported rates.” What are these rates?

Supported rates: 1, 2, 5.5, 11

```
‣ Frame 1: 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
‣ Radiotap Header v0, Length 24
‣ 802.11 radio information
‣ IEEE 802.11 Beacon frame, Flags: .....C
‣ IEEE 802.11 Wireless Management
  ‣ Fixed parameters (12 bytes)
  ‣ Tagged parameters (119 bytes)
    ‣ Tag: SSID parameter set: "30 Munroe St"
    ‣ Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), [Mbit/sec]
      Tag Number: Supported Rates (1)
      Tag length: 4
      Supported Rates: 1(B) (0x82)
      Supported Rates: 2(B) (0x84)
      Supported Rates: 5.5(B) (0x8b)
      Supported Rates: 11(B) (0x96)
    ‣ Tag: DS Parameter set: Current Channel: 6
    ‣ Tag: Traffic Indication Map (TIM): DTIM 0 of 1 bitmap
    ‣ Tag: Country Information: Country Code US, Environment Indoor
    ‣ Tag: EDCA Parameter Set
    ‣ Tag: ERP Information
    ‣ Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54, [Mbit/sec]
    ‣ Tag: Vendor Specific: Airgo Networks, Inc.
    ‣ Tag: Vendor Specific: Microsoft Corp.: WMM/WME: Parameter Element
```

Question 6: What are the “extended supported rates”?

Extended support rates: 6, 9, 12, 18, 24, 36, 48, 54

```
▶ Frame 1: 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
▶ Radiotap Header v0, Length 24
▶ 802.11 radio information
▶ IEEE 802.11 Beacon frame, Flags: .....C
▼ IEEE 802.11 Wireless Management
  ▶ Fixed parameters (12 bytes)
  ▼ Tagged parameters (119 bytes)
    ▶ Tag: SSID parameter set: "30 Munroe St"
    ▶ Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B), [Mbit/sec]
    ▶ Tag: DS Parameter set: Current Channel: 6
    ▶ Tag: Traffic Indication Map (TIM): DTIM 0 of 1 bitmap
    ▶ Tag: Country Information: Country Code US, Environment Indoor
    ▶ Tag: EDCA Parameter Set
    ▶ Tag: ERP Information
    ▼ Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54, [Mbit/sec]
      Tag Number: Extended Supported Rates (50)
      Tag length: 8
      Extended Supported Rates: 6(B) (0x8c)
      Extended Supported Rates: 9 (0x12)
      Extended Supported Rates: 12(B) (0x98)
      Extended Supported Rates: 18 (0x24)
      Extended Supported Rates: 24(B) (0xb0)
      Extended Supported Rates: 36 (0x48)
      Extended Supported Rates: 48 (0x60)
      Extended Supported Rates: 54 (0x6c)
    ▶ Tag: Vendor Specific: Airgo Networks, Inc.
    ▶ Tag: Vendor Specific: Microsoft Corp.: WMM/WME: Parameter Element
```

Question 7: What is the channel number?

Channel number: 6

```
▶ Frame 1: 183 bytes on wire (1464 bits), 183 bytes captured (1464 bits)
▶ Radiotap Header v0, Length 24
▼ 802.11 radio information
  PHY type: 802.11b (HR/DSSS) (4)
  Short preamble: False
  Data rate: 1.0 Mb/s
  Channel: 6
  Frequency: 2437MHz
  Signal strength (dB): 71 dB
  Signal strength (dBm): -29 dBm
  Noise level (dBm): -100 dBm
  Signal/noise ratio (dB): 71 dB
  ▶ [Duration: 1464µs]
▶ IEEE 802.11 Beacon frame, Flags: .....C
▶ IEEE 802.11 Wireless Management
```

Question 8: How many “Association Request” are displayed?

Number of displayed “association request” packets: 17

wlan.fc.type_subtype == 0x00					
No.	Time	Source	Destination	Protocol	Length Info
1227	33.079714	d1:b6:4f:00:16:b6	MS-NLB-PhysServer-3...	802.11	111 Association Request, SN=3775, FN=4, Flags=.pm...F.C
1750	49.651078	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1607, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1751	49.653218	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1607, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1824	53.789944	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1613, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1825	53.790943	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1613, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1827	53.793568	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1613, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1926	57.903699	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1927	57.904945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1932	57.911195	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1933	57.915945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1934	57.924199	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1935	57.936216	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
1937	57.939196	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1620, FN=0, Flags=.....C, SSID="linksys_SES_24086"
2126	62.176945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1645, FN=0, Flags=.....C, SSID="linksys_SES_24086"
2127	62.178194	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	107 Association Request, SN=1645, FN=0, Flags=.....C, SSID="linksys_SES_24086"
2162	63.169910	Intel_d1:b6:4f	CiscoLinksys_f7:1d:...	802.11	89 Association Request, SN=1648, FN=0, Flags=.....C, SSID="30 Munroe St"
2307	70.179949	CiscoLinksys_f5:ba:7b	f9:ff:ff:ff:ff:ff	802.11	132 Fragmented IEEE 802.11 frame

Question 9: Which AP sent a “Association Response” to “Association Request”?

Access point that sent an association response: CiscoLinksys

wlan.fc.type_subtype == 0x01					
No.	Time	Source	Destination	Protocol	Length Info
2166	63.192101	CiscoLinksys_f7:1d:51	Intel_d1:b6:4f	802.11	94 Association Response, SN=3728, FN=0, Flags=.....C

Question 10: What is the IP address obtained by the client station?

IP address obtained by the client station: 192.168.1.109

bootp.option.type == 53 && bootp.option.value == 5					
No.	Time	Source	Destination	Protocol	Length Info
2211	66.217462	192.168.1.1	192.168.1.109	DHCP	638 DHCP ACK - Transaction ID 0x2733a47c

User Datagram Protocol, Src Port: 67, Dst Port: 68	
Dynamic Host Configuration Protocol (ACK)	
Message type: Boot Reply (2)	
Hardware type: Ethernet (0x01)	
Hardware address length: 6	
Hops: 0	
Transaction ID: 0x2733a47c	
Seconds elapsed: 0	
Bootp flags: 0x0000 (Unicast)	
Client IP address: 0.0.0.0	
Your (client) IP address: 192.168.1.109	
Next server IP address: 0.0.0.0	
Relay agent IP address: 0.0.0.0	
Client MAC address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)	
Client hardware address padding: 00000000000000000000	
Server host name not given	
Boot file name not given	
Magic cookie: DHCP	
Option: (53) DHCP Message Type (ACK)	
Option: (54) DHCP Server Identifier (192.168.1.1)	
Option: (51) IP Address Lease Time	
Option: (1) Subnet Mask (255.255.255.0)	
Option: (3) Router	
Option: (6) Domain Name Server	
Option: (15) Domain Name	
Option: (255) End	

Question 11: What is the DHCP server IP address?

DHCP server IP address: 192.168.1.1

[illegible]

Question 12: What is the “IP address leased time”?

IP address leased time: 1 day

```

▼ Dynamic Host Configuration Protocol (ACK)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x2733a47c
  Seconds elapsed: 0
  ▶ Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 192.168.1.109
  Next server IP address: 0.0.0.0
  Relay agent IP address: 0.0.0.0
  Client MAC address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)
  Client hardware address padding: 000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  ▶ Option: (53) DHCP Message Type (ACK)
  ▶ Option: (54) DHCP Server Identifier (192.168.1.1)
  ▼ Option: (51) IP Address Lease Time
    Length: 4
    IP Address Lease Time: 1 day (86400)
  ▶ Option: (1) Subnet Mask (255.255.255.0)
  ▶ Option: (3) Router
  ▶ Option: (6) Domain Name Server
  ▶ Option: (15) Domain Name

```

Question 13: Locate packet number 480, What is the HTTP request?

HTTP request: GET request

```
▶ Frame 480: 537 bytes on wire (4296 bits), 537 bytes captured (4296 bits)
▶ Radiotap Header v0, Length 24
▶ 802.11 radio information
▶ IEEE 802.11 QoS Data, Flags: .....TC
▶ Logical-Link Control
▶ Internet Protocol Version 4, Src: 192.168.1.109, Dst: 128.119.245.12
▶ Transmission Control Protocol, Src Port: 2538, Dst Port: 80, Seq: 1, Ack: 1, Len: 435
▼ Hypertext Transfer Protocol
  ▶ GET /wireshark-labs/alice.txt HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.8.0.12) Gecko/20070508 Firefox/1.5.0.12\r\n
    Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,image/png,*/*;q=0.5\r\n
    Accept-Language: en-us,en;q=0.5\r\n
    Accept-Encoding: gzip,deflate\r\n
    Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7\r\n
    Keep-Alive: 300\r\n
    Connection: keep-alive\r\n
    \r\n
    [Response in frame: 868]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/alice.txt]
```

Question 14: What is the name of the file downloaded in Question 13?

Name of file downloaded: alice.txt

```
▶ Frame 480: 537 bytes on wire (4296 bits), 537 bytes captured (4296 bits)
▶ Radiotap Header v0, Length 24
▶ 802.11 radio information
▶ IEEE 802.11 QoS Data, Flags: .....TC
▶ Logical-Link Control
▶ Internet Protocol Version 4, Src: 192.168.1.109, Dst: 128.119.245.12
▶ Transmission Control Protocol, Src Port: 2538, Dst Port: 80, Seq: 1, Ack: 1, Len: 435
▼ Hypertext Transfer Protocol
  ▶ GET /wireshark-labs/alice.txt HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.8.0.12) Gecko/20070508 Firefox/1.5.0.12\r\n
    Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,image/png,*/*;q=0.5\r\n
    Accept-Language: en-us,en;q=0.5\r\n
    Accept-Encoding: gzip,deflate\r\n
    Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7\r\n
    Keep-Alive: 300\r\n
    Connection: keep-alive\r\n
    \r\n
    [Response in frame: 868]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/alice.txt]
```

Question 15: Extract the file in Question 14 from the PCAP file

Please see `alice.txt` in submission.